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NPS/001/025 – Technical Specification for Cable Guards

1. Purpose

The purpose of this document is to detail the technical requirements for cable guards required for the protection of cables fitted to poles and buildings on the CE Electric UK distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

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2. Scope

This document applies to guards and earth lead casings designed to protect cables from inadvertent damage or vandalism. BS EN 50086 - 1: 1993 the Specification for Conduit systems for cable management – General requirements, has been adopted as a standard for the product range as no specific standard exists.

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3. General

3.1 Application

Cable guards and earth lead casings are used to protect conductors and cables attached to poles from inadvertent damage or vandalism. An anti-vandal earth lead casing is required for areas prone to high levels of vandalism or cable theft. It shall exceed the impact and mechanical properties of the standard guards.

Cable guards and earth lead casings shall be designed for attachment to wood and steel poles with nails, screws or steel 'bandit' tape.

3.2 Construction

Product dimensions shall be as detailed in CE Electric UK drawings: -

- 1) 1.09.101.0149 Sheet 3 – Earth Lead Casing.
- 2) 1.09.101.0171 Sheet 1 – Cable Guards for Wood and Steel Poles

The list of items covered by this specification is detailed in Appendix 2.

They shall be free from sharp edges, burrs or surface projections which are likely to damage insulated cables or inflict injury to the installer.

They shall be manufactured from PVC, heavy duty polyethylene or similar material that is UV stable, creosote resistant and as required to conform to this specification

3.3 Classification

BS EN 50086 – 1 provides classification options for a number of technical properties. Several of these key properties are listed below and should be used as a guide. Appendix A shall be completed by the supplier with confirmation and comments regarding compliance with this or alternative specifications.

Properties	Level	BS EN 50086 - Code
Resistance to compression	Heavy compression strength	4
Resistance to impact	Heavy impact strength	4
Minimum permanent application and installation temperature	- 15° C	3
Maximum permanent application and installation temperature	+ 60° C	1
Resistance to bending	Pliable/Self-recovering	3
Electrical properties	With electrical insulating properties	2
Resistance to ingress of solid objects	Protect against solid foreign objects of 2.5mm dia. and greater	3
Resistance to ingress of moisture	Not declared	0
Corrosion resistance of metallic and composite conduit systems	Low protection inside and out	1
Tensile strength	None declared	0
Resistance to flame propagation	Flame propagating	2
Suspended load capacity	None declared	0

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4. References

4.1 External Documentation

<u>Reference</u>	<u>Title</u>
BS EN 50086; Part 1: 1994	Specification for conduit systems for cable management – general requirements

4.2 Internal documentation

<u>Reference</u>	<u>Title</u>
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5. Definitions

<u>Term</u>	<u>Definition</u>
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6. Authority for issue

Written by:	Steven Salkeld	Standards Engineer	17/11/2009	Steven Salkeld
Technically assured by:	Richard McMahon	Standards Engineer	17/11/2009	Richard McMahon
CDS assured by:	Sean Johnson	CDS Administrator	30/11/2009	Sean Johnson
Authorised for Publication by:	Chris Holdsworth	Standards Manager	18/11/2009	Chris Holdsworth

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Appendix 1 – Supplier Self Certification Compliance Declaration

No national or international standard exists for this product type therefore BS EN 50086 has been adopted to benchmark the performance criteria. Suppliers are requested to complete the table below and provide comments on compliance with the specification. An additional table shall be completed if more than one product type is required.

Properties	Comment/Test Evidence
Resistance to compression	
Resistance to impact	
Minimum permanent application and installation temperature	
Maximum permanent application and installation temperature	
Resistance to bending	
Electrical properties	
Resistance to ingress of solid objects	
Corrosion resistance of metallic and composite conduit systems	
Resistance to flame propagation	
Dimensional requirements	
UV Stability	

*Where appropriate type test reports shall be submitted to support any statements provided in the table above.

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Appendix 2 – Schedule of Items

The following items are currently stocked by CE Electric UK but the opportunity to harmonise may be considered if suitable offers are received.

252492 - Type 1 - 40mm internal diameter x 3300mm
252469 - Type 2 - 50mm internal diameter x 3300mm
224642 - Type 2 Extension Piece - 50mm internal diameter x 400mm
252505 - Type 3 - 75mm internal diameter x 3300mm
252454 - Type 4 - 100mm internal diameter x 3300mm
235548 - Cable Earth Lead Casing – 20mm internal diameter x 3300mm – Standard Type
235550 - Cable Earth Lead Casing – 20mm internal diameter x 3300mm – Anti-vandal Type